

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
Parent Process
Parent PID = 970

Child PID = 971
Child Process Created
Child PID = 971
Parent PID = 970
Executing 'ls' command...
total 60
-rw-r--r-- 1 suresh suresh 0 Sep 2 03:22 LICENSE
-rw-r--r-- 1 suresh suresh 0 Sep 2 03:22 Makefile
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:41 Practical
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:40 Prctical
-rw-r--r-- 1 suresh suresh 0 Sep 2 03:22 README.md
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 assets
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:13 bin
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:18 docs
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:16 include
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:13 obj
-rwxr-xr-x 1 suresh suresh 16264 Sep 2 03:31 process
-rw-r--r-- 1 suresh suresh 997 Sep 2 03:30 process.c
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 scripts
drwxr-xr-x 2 suresh suresh 4096 Sep 2 03:20 src
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 tests

Child process completed.
suresh@DESKTOP-QCI8C9M:~/OSSP$ nano shell.c
suresh@DESKTOP-QCI8C9M:~/OSSP$ gcc shell.c -o shell
suresh@DESKTOP-QCI8C9M:~/OSSP$ ./shell
myshell> hello
You entered: hello
myshell> is
You entered: is
myshell> pwd
You entered: pwd
myshell> exit
Exiting shell...
suresh@DESKTOP-QCI8C9M:~/OSSP$
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

```
suresh@DESKTOP-QCI8C9M: ~/OSSP
Parent PID = 970
Executing 'ls' command...
total 60
-rw-r--r-- 1 suresh suresh 0 Sep  2 03:22 LICENSE
-rw-r--r-- 1 suresh suresh 0 Sep  2 03:22 Makefile
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:41 Practical
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drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 assets
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drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:18 docs
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:16 include
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-rwxr-xr-x 1 suresh suresh 16264 Sep  2 03:31 process
-rw-r--r-- 1 suresh suresh 997 Sep  2 03:30 process.c
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 scripts
drwxr-xr-x 2 suresh suresh 4096 Sep  2 03:20 src
drwxr-xr-x 2 suresh suresh 4096 Aug 14 05:19 tests

Child process completed.
suresh@DESKTOP-QCI8C9M:~/OSSP$ nano shell.c
suresh@DESKTOP-QCI8C9M:~/OSSP$ gcc shell.c -o shell
suresh@DESKTOP-QCI8C9M:~/OSSP$ ./shell
myshell> hello
You entered: hello
myshell> is
You entered: is
myshell> pwd
You entered: pwd
myshell> exit
Exiting shell...
suresh@DESKTOP-QCI8C9M:~/OSSP$ nano copyfile.c
suresh@DESKTOP-QCI8C9M:~/OSSP$ nano copyfile.c
suresh@DESKTOP-QCI8C9M:~/OSSP$ nano File1.txt
suresh@DESKTOP-QCI8C9M:~/OSSP$ gcc copyfile.c -o copyfile
suresh@DESKTOP-QCI8C9M:~/OSSP$ ./copyfile
File copied successfully!
suresh@DESKTOP-QCI8C9M:~/OSSP$ cat File2.txt
Hello World!
This is File1.
This content will be copied to File2.
suresh@DESKTOP-QCI8C9M:~/OSSP$
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    ssize_t n;

    // Open File1.txt for reading
    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0)
    {
        perror("Error opening File1.txt");
        return 1;
    }

    // Open File2.txt for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0)
    {
        perror("Error opening File2.txt");
        close(fd1);
        return 1;
    }
}
```

```
// Read from File1 and write to File2
while ((n = read(fd1, buffer, sizeof(buffer))) > 0)
{
    if (write(fd2, buffer, n) != n)
    {
        perror("Error writing to File2.txt");
        close(fd1);
        close(fd2);
        return 1;
    }
}

if (n < 0)
{
    perror("Error reading File1.txt");
}
else
{
    printf("File copied successfully.\n");
}

// Close both files
close(fd1);
close(fd2);

return 0;
}
```